the factors involved in the process of production.

- ***Risk-taking and Uncertainty-bearing:*** There are risk-taking and uncertainty-functions for an entrepreneur bearing obstacles. Risks may be insured but uncertainties cannot be insured. They reduce the profit.

## 3.3 Production Function

Production function refers to the relationship among units of the factors of production (inputs) and the resultant quantity of a good produced (output).

According to George J. Stigler, *"Production function is the relationship between inputs of productive services per unit of time and outputs of product per unit of time."*



George J.Stigler

Production function may be expressed as: Q = f (N, L, K, T) Where, Q = Quantity of output, N = Land; L = Labour; K = Capital; and T = Technology. Depending on the efficiency of the producer, this production function varies.

The function implies that the level of output (Q) depends on the quantities of different inputs (N, L, K, T) available to the firm.

**Short-run Production and Long run Production**

In Micro economics, the distinction between long run and short run is made on the basis of fixed inputs that inhibit the production.

The short-run is the period where some inputs are variable, while others are fixed. Another feature is that new firms do not enter into the industry and existing firms may not leave the industry.

Long run, on the other hand, is the period featured by the entry of new firms to the industry and the exit of existing firms from the industry.

In general, Production function may be classified into two

- Short-run Production Function as illustrated by the Law of Variable Proportions.
- Long-run Production Function as explained by the Laws of Returns to Scale.

## 3.4 Law of Variable Proportions

The law states that if all other factors are fixed and one input is varied in the short run, the total output will increase at an increasing rate at first instance, be constant at a point and then eventually decrease. Marginal product will become negative at last.

According to G.Stigler, *"As equal increments of one input are added, the inputs of other productive services being held constant, beyond a certain point, the resulting increments of product will decrease, i.e., the marginal product will diminish".*

### Assumptions

The Law of Variable Proportions is based on the following assumptions.



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